

## Three species of fishes from an eutrophic, seasonally alkaline lake are not more tolerant to acute exposure to high pH in the laboratory

D. M. SCOTT\* AND R. W. WILSON

*School of Biosciences, Hatherly Laboratories, University of Exeter,  
Exeter, EX4 4PS, U.K.*

*(Received 17 August 2005, Accepted 9 October 2006)*

A number of different freshwater fish species (perch *Perca fluviatilis*, roach *Rutilus rutilus* and rudd *Scardinius erythrophthalmus*) from either eutrophic (Slapton Ley, a seasonally alkaline lake) or non-eutrophic waters were compared with respect to their sodium uptake kinetics and tolerance to acute (1 h) exposure to pH 9.5. Further comparisons were made with rainbow trout *Oncorhynchus mykiss* and brown trout *Salmo trutta*. The influence of fish size was also investigated in rainbow trout. Exposure to pH 9.5 was found to disrupt sodium balance and inhibit ammonia excretion in all species and sizes of fishes. The origin of fishes did not have a significant effect on the sodium uptake kinetics or the physiological responses to high pH water. The fishes from the eutrophic lake therefore did not appear to have any increased tolerance to acute exposure to alkaline water. In contrast to previous studies there was no inhibition of Na<sup>+</sup> uptake during exposure to high pH. Indeed in some groups of fish Na<sup>+</sup> uptake was actually stimulated, as was Na<sup>+</sup> efflux. These differences are attributed to experimental water composition and interspecific differences in physiology. It was not always possible to size-match fishes of the different species, so rainbow trout were used to assess the effect of body mass (from 2 to 40 g), on Na<sup>+</sup> uptake kinetics and Na<sup>+</sup> or ammonia fluxes during alkaline water exposure in rainbow trout. Size had no significant effect on these measurements within this narrow range, which helps validate the comparison between species in this study.

© 2007 The Fisheries Society of the British Isles

**Key words:** ammonia excretion; body size; eutrophic; gills; ion transport; sodium uptake kinetics.

### INTRODUCTION

Exposure to high pH water is known to cause substantial impairment of various functions of the gills in freshwater fishes including ion and acid-base regulation and excretion of nitrogenous waste. Net losses of Na<sup>+</sup> and Cl<sup>-</sup> have been recorded in rainbow trout *Oncorhynchus mykiss* (Walbaum) and attributed to a reduction in branchial uptake rates and not increased diffusive efflux (Wright & Wood, 1985; Wilkie & Wood, 1991, 1994; Wilkie *et al.*, 1996,

\*Author to whom correspondence should be addressed. Tel.: +44 (0)1392 264652; fax: +44 (0)1392 263700; email: [d.m.scott@exeter.ac.uk](mailto:d.m.scott@exeter.ac.uk)

1999; Laurent *et al.*, 2000). Fishes also become alkalotic and the internal counter-ions for  $\text{Na}^+$  and  $\text{Cl}^-$  uptake ( $\text{H}^+$  and  $\text{HCO}_3^-$ , respectively) are reduced, which is thought to be the cause of the reduced ion uptake (Goss & Wood, 1991; Wilkie *et al.*, 1999). Immediate inhibition of ammonia excretion and elevation of plasma ammonia have also been observed in rainbow trout during exposure to alkaline water (Wilkie & Wood, 1991, 1994; Wilkie *et al.*, 1996; Wilson *et al.*, 1998; Laurent *et al.*, 2000). This is attributed to a reduction in the ammonia partial pressure gradient across the gills, which is thought to be the most important factor driving ammonia excretion, due to a reduction in  $\text{H}^+$  available to trap  $\text{NH}_3$  as  $\text{NH}_4^+$  in the gill boundary layer (Wright *et al.*, 1989; Wilson *et al.*, 1994).

Despite the above understanding of how high pH acts upon the physiology of freshwater fishes, there are several important factors that may influence toxicity for which there is no information. These include variability between species, the potential for acclimation or adaptation to increase tolerance and body size. The vast majority of the information on the physiological responses of fishes to alkaline pH is based on salmonids, in particular the rainbow trout. There are clear differences, however, between freshwater species with respect to their mechanisms for regulating ions, acid-base balance and nitrogenous waste excretion (Hughes, 1966; Goss *et al.*, 1992; Gonzalez *et al.*, 2002; Wood *et al.*, 2002; Marshall & Grosell, 2005). Such differences between species include the gill ionic permeability, the affinity and capacity of ion uptake transport systems, the types and sensitivity of the transporters involved and the flexibility of the nitrogenous waste excretion system. Given that alkaline pH targets the gill transport processes, it may therefore be inappropriate to rely solely upon studies of rainbow trout when exploring the potential for alkaline pH to impact upon other species. Furthermore, fish populations that live in eutrophic bodies of water may experience regular periods of alkaline pH, *e.g.* annually during summer blooms of algae or macrophytes (Paerl *et al.*, 2001). This could increase the alkaline tolerance of these populations through either acclimatory changes during individual lifetimes or selection over multiple generations.

Size is one of the most important features of living organisms that influences countless aspects of biology and function (Schmidt-Nielsen, 1984). The allometric relationship between mass-specific metabolic rate and body size in resting animals is well established for homeotherms, but is also true for poikilotherms (including fishes) once variations due to body temperature are taken into account (Randall *et al.*, 2002). Body size has been found to be very important when considering the sensitivity of aquatic animals to toxicants that, like alkaline pH, target ion transport processes at the gill (*e.g.* silver and copper). For these toxicants smaller fishes and invertebrates were found to be more sensitive, due to their higher mass-specific ion turnover rates (uptake and loss) so that the same degree of inhibition of ion uptake results in a faster depletion of internal ion stores (Bianchini *et al.*, 2002; Grosell *et al.*, 2002). Thus it might be predicted that the effects of high pH on fishes would also be size-dependent.

The primary aim of the present study was to evaluate the tolerance to high pH of three species of fishes from a seasonally alkaline lake, thereby assessing their potential acclimation to these conditions. Three species [perch *Perca fluviatilis* L., roach *Rutilus rutilus* (L.) and rudd *Scardinius erythrophthalmus* (L.)] were

chosen that inhabit Slapton Ley (50°17' N; 3°39' W), a freshwater lake that has become increasingly eutrophic since the 1950s (Johnes & Heathwaite, 1997). Slapton Ley is known to experience annual alkalization regularly reaching pH 9.0–9.5 for 3 months each summer (Scott, 2003; Scott *et al.*, 2005). Comparisons were made with fishes of the same species from non-eutrophic waters as well as with the more commonly studied salmonids: rainbow trout and brown trout *Salmo trutta* L. The comparison between species included a quantitative assessment of the effect of an acute exposure to pH 9.5 on ion fluxes and ammonia excretion, as well as characterizing the affinity ( $K_m$ ) and capacity ( $V_{max}$ ) for  $Na^+$  uptake. In addition the potential for body mass to influence the effects of high pH, using three different sizes of rainbow trout, was examined. This was important in order to help with the interpretation of the data, where it was not always possible to obtain size-matched groups of fishes.

## MATERIALS AND METHODS

### EXPERIMENTAL ANIMALS

Physiological experiments were conducted throughout the period January 2001 to March 2003. During this time perch ( $1.5 \pm 0.1$  g, mean  $\pm$  s.e.), roach ( $7.7 \pm 0.4$  g) and rudd ( $19.7 \pm 1.3$  g) were caught from Slapton Ley (average composition, means  $\pm$  s.e.:  $[Na^+] = 688 \pm 15$ ,  $[K^+] = 80 \pm 3$ ,  $[Ca^{2+}] = 586 \pm 30$ ,  $[Mg^{2+}] = 320 \pm 15$   $\mu\text{mol l}^{-1}$ , pH range 7.2–10.3) by electrofishing during times when the lake was of neutral pH. For comparison rainbow trout ( $14.9 \pm 0.6$  g for acute pH and  $Na^+$  uptake kinetics experiments and  $2.0 \pm 0.1$  g,  $10.9 \pm 0.7$  g,  $41.8 \pm 0.3$  g for size comparison experiments), brown trout ( $21.8 \pm 0.2$  g), rudd ( $50.4 \pm 3.2$  g), perch ( $5.2 \pm 0.3$  g) and roach ( $5.9 \pm 0.2$  g) were obtained from local spring fed fish farms, which had no record of problems with changes in water pH. Fishes were maintained in chilled dechlorinated Exeter tap water (average composition:  $[Na^+] = 412 \pm 15$ ,  $[K^+] = 55 \pm 3$ ,  $[Ca^{2+}] = 624 \pm 18$ ,  $[Mg^{2+}] = 139 \pm 7$ , titratable alkalinity (TAlk) =  $969 \pm 42$   $\mu\text{mol l}^{-1}$ , pH =  $7.46 \pm 0.04$ , temperature =  $10.8 \pm 0.2^\circ\text{C}$ ) and fed with commercial fish food and live chironomid larvae from in-house cultures. Photoperiod was 12L : 12D. Fishes were allowed at least 2 weeks to recover from capture and transportation before being used in a physiological experiment.

### EXPERIMENTAL PROTOCOL

The system used for conducting flux experiments consisted of a re-circulating 150 l volume of temperature-controlled water ( $10.0^\circ\text{C}$ , range  $\pm 0.2^\circ\text{C}$ ), which fed a number of flux chambers in which individual fishes were placed. Flux chambers were darkened, plastic containers supplied with aerated water at a flow rate of  $350 \text{ ml min}^{-1}$  from a header tank. The flux chambers were of variable size, designed to minimize the ratio of fish mass (g) to water volume (ml) (ratios ranged between 1:9 and 1:35) and hence maximize the sensitivity of the flux measurements. Each chamber had its own air supply, which also aided mixing during fluxes. Food was withheld from fishes during the week prior to an experiment in order to allow ammonia excretion rates to stabilize (Brett & Zala, 1975). Twenty four hours prior to an experiment individual fishes were transferred into flux chambers in order to allow metabolic and ionic fluxes to stabilize following the stress of handling.

Each flux consisted of a period of time (nominally 1 h) in which the chambers were sealed and flow of water stopped. At the beginning of the flux period an aliquot of  $^{22}\text{NaCl}$  was added to the chamber water ( $0.0074$ – $0.0370$  KBq, depending on the

non-radioactive  $\text{Na}^+$  concentration) to allow measurement of unidirectional  $\text{Na}^+$  fluxes. Varying the concentration of  $^{22}\text{NaCl}$  prevented complications from increased internal specific activity within the fishes and the associated backflux of radioactivity (Maetz, 1956). Following a 5 min mixing period an initial 15 ml water sample was removed, followed by a final water sample taken at the end of the flux period. The chamber was then flushed out in order to remove any radioactivity before reconnection to the re-circulating system. The mass of the experimental animals was then recorded to the nearest 0.01 g.

## ANALYTICAL TECHNIQUES AND FLUX CALCULATIONS

Ion concentrations ( $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{K}^+$  and  $\text{Na}^+$ ) were measured using atomic absorption spectrophotometry (Pye Unicam SP9) or flame photometry (Corning 410). Total ammonia ( $T_{\text{Amm}}$ ) in water samples was measured in triplicate using a micro-modification of the salicylate-hypochlorite assay (Verdouw *et al.*, 1978). Triplicate 2 ml water samples were analysed for  $^{22}\text{Na}$  activity, counted on a Packard Cobra auto-gamma counter (B5002). Water pH was measured using a low conductivity pH electrode (Russell KCET/11) and Hanna meter (HI 8710), and TALK was determined as described by McDonald & Wood (1981), in which water samples were titrated to a fixed end point (pH 4.0) with  $0.02 \text{ mol l}^{-1} \text{ HCl}$ . Unidirectional  $\text{Na}^+$  flux and ammonia excretion rates were calculated as described by Wright & Wood (1985).

## ACUTE EXPOSURE TO HIGH pH—SODIUM AND AMMONIA FLUXES

To investigate the tolerance of fishes to an acute pH 9.5 exposure, with respect to disruption of  $\text{Na}^+$  balance and ammonia excretion, 10 fish of each species or source were used per experiment. Each experiment consisted of three successive flux periods (control, experimental and recovery) of 1 h duration with 0.5 h in between when the chambers were flushed with clean water. The control and recovery media consisted of dechlorinated Exeter tap water at pH *c.* 7.5. An average pH of 9.5, range  $\pm 0.1$  was maintained during the experimental flux period by manual checking every 10 min and appropriate addition of  $0.1 \text{ mol l}^{-1} \text{ KOH}$ . Due to the addition of KOH the chamber water  $\text{K}^+$  concentrations varied during the experimental flux periods, but never exceeded  $1.4 \text{ mmol l}^{-1}$ , a value substantially below that which is thought to be toxic to fishes (Wilkie *et al.*, 1993).

## SODIUM UPTAKE KINETICS

In order to make a further comparison between Slapton and non-Slapton fishes a series of fluxes was conducted to measure the  $\text{Na}^+$  uptake kinetics. Each  $\text{Na}^+$  kinetics experiment consisted of six successive flux periods in which the  $\text{Na}^+$  concentration within the chamber water was progressively increased (ranging between 10 and  $1500 \text{ } \mu\text{mol l}^{-1}$ ), by addition of aliquots of NaCl. The  $\text{Na}^+$  concentration was approximately doubled for each successive flux, with the starting  $\text{Na}^+$  concentration varying between 10 and  $40 \text{ } \mu\text{mol l}^{-1}$  depending on the species. The six fluxes were carried out consecutively, with partial replacement of chamber water at the end of each flux in order to maintain ammonia levels in the chambers  $< 10 \text{ } \mu\text{mol l}^{-1}$ . Each flux lasted 0.75 h except the last flux (highest  $\text{Na}^+$  concentration), which was 1 h in duration.

At the beginning of each experiment chambers were first flushed through with  $\text{Na}^+$ -free water, which was made using salts that duplicated the  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  and TALK composition of Exeter tap water as closely as possible. The  $\text{Na}^+$ -free water ( $[\text{Na}^+] = 2.2 \pm 0.3$ ,  $[\text{Ca}^{2+}] = 529 \pm 25$ ,  $[\text{Mg}^{2+}] = 158 \pm 9$ , TALK =  $1479.0 \pm 0.1 \text{ } \mu\text{mol l}^{-1}$ , pH =  $7.55 \pm 0.09$ , temperature =  $10.6 \pm 0.2^\circ \text{C}$ ) was made by adding the appropriate amounts of  $\text{CaCO}_3$  and  $(\text{MgCO}_3)_4 \cdot \text{Mg}(\text{OH})_2 \cdot 5\text{H}_2\text{O}$  salts to deionized water. Because these salts

are relatively insoluble in water at neutral pH, the water was bubbled with 100% CO<sub>2</sub> overnight to achieve complete solution. The water was then vigorously aerated for 24 h to drive off the CO<sub>2</sub>. The resulting medium had appropriate levels of Ca<sup>2+</sup>, Mg<sup>2+</sup> and TALK but negligible Na<sup>+</sup> (Goss & Wood, 1990). Changing water Na<sup>+</sup> concentrations revealed distinctive saturation kinetics for Na<sup>+</sup> uptake, and  $K_m$  and  $V_{max}$  values were determined for individual fishes by fitting data to the Michaelis–Menten equation ( $\text{Na}^+ \text{ uptake} = V_{max} \times [\text{water Na}^+] / (K_m + [\text{water Na}^+])$ ) using non-linear regression (SigmaStat).

A series of experiments were conducted with rainbow trout of three different sizes; categorized as small ( $2.0 \pm 0.1$  g), medium ( $10.9 \pm 0.7$  g) and large ( $41.8 \pm 0.3$  g), which covers the range of all the fishes used within the present study. High pH and Na<sup>+</sup> kinetics experiments were carried out for each size group of rainbow trout as described above.

## STATISTICS

Where the Na<sup>+</sup> flux values and ammonia excretion rates were normally distributed they were compared using one-way repeated measures ANOVA ( $P \leq 0.05$ ) (Tukey's test) with each fish as its own control. Normality was checked using the Kolmogorov–Smirnov test and when normality failed values were compared using repeated measures ANOVA on ranks (Dunnnett's method) with each fish as its own control. The per cent decreases in ammonia excretion for the different fish groups were first arcsine transformed and then compared using one-way ANOVA ( $P \leq 0.05$ ) followed by Tukey's test. The kinetic parameters  $K_m$  and  $V_{max}$  were compared using one-way ANOVA unless there was failure of the equal variance test (Levene Median test) in which case Kruskal–Wallis one-way ANOVA on ranks ( $P \leq 0.05$ ) with pair-wise multiple comparison procedures (Dunn's method) was used. Relationships between size of rainbow trout and Na<sup>+</sup> and ammonia fluxes,  $K_m$  and  $V_{max}$  values were tested using Spearman rank ( $P \leq 0.05$ ). All experiments were conducted under a Home Office licence and with approval of the Ethical Review Committee of the University of Exeter.

## RESULTS

The results of the Na<sup>+</sup> flux analysis for the different fish groups before, during and after exposure to pH 9.5 water are shown in Figs 1 and 2. Direct comparisons of the Na<sup>+</sup> flux values of the different fish groups are not possible because each group has different control values. The Na<sup>+</sup> fluxes of the different fish groups have been compared with respect to the changes from their own control values. The Slapton and non-Slapton roach showed different responses to high pH with respect to Na<sup>+</sup> uptake ( $J_{Na}^{in}$ ), but both groups experienced a significant increase in Na<sup>+</sup> efflux ( $J_{Na}^{out}$ ; 2.9-fold increase) and more negative net Na<sup>+</sup> balance ( $J_{Na}^{net}$ ) during the high pH exposure. Slapton roach failed to recover  $J_{Na}^{net}$  completely, whilst the non-Slapton roach did recover. Slapton and non-Slapton rudd showed very similar patterns and had virtually no response to the acute high pH exposure. Both groups experienced significant and similar decreases in  $J_{Na}^{in}$  and  $J_{Na}^{out}$  during the recovery period with  $J_{Na}^{net}$  closer to zero, compared to the controls.

The only significant change to  $J_{Na}^{in}$  in the perch groups was a decrease during the recovery period within the Slapton fish. Both Slapton and non-Slapton perch showed a significant increase in  $J_{Na}^{out}$  (2.3 and 5.0-fold respectively) and more negative  $J_{Na}^{net}$  during pH 9.5 exposure. Both groups failed to fully recover net Na<sup>+</sup> balance to the control values during the recovery period. Furthermore,

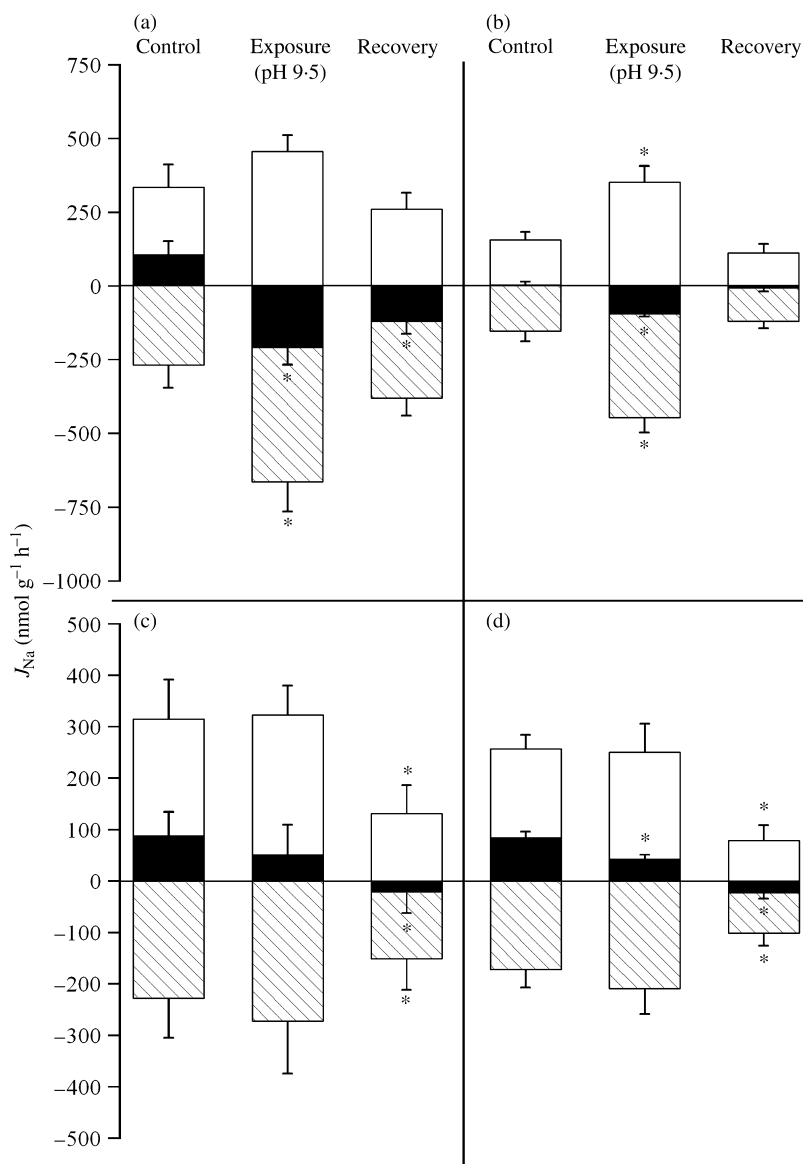


FIG. 1. Sodium fluxes [ $J_{Na}^{\text{in}}$  (□),  $J_{Na}^{\text{net}}$  (■) and  $J_{Na}^{\text{out}}$  (▨)] in (a) Slapton and (b) non-Slapton roach and (c) Slapton and (d) non-Slapton rudd before, during and after exposure to pH 9.5 water. Mean  $\pm$  s.e. values are shown with negative and positive values indicating a loss from or gain to the fish, respectively. \*, a significant difference ( $P < 0.05$ ) from the corresponding control value.

$J_{Na}^{\text{out}}$  did not recover in the non-Slapton perch during the recovery period. Neither rainbow trout nor brown trout showed any significant changes to  $J_{Na}^{\text{net}}$  during these experiments. The rainbow trout experienced a significant elevation in  $J_{Na}^{\text{in}}$  (2.0-fold) and  $J_{Na}^{\text{out}}$  (1.8-fold) during high pH exposure, followed by a significant decrease in  $J_{Na}^{\text{in}}$  in the recovery period. In the brown trout the only

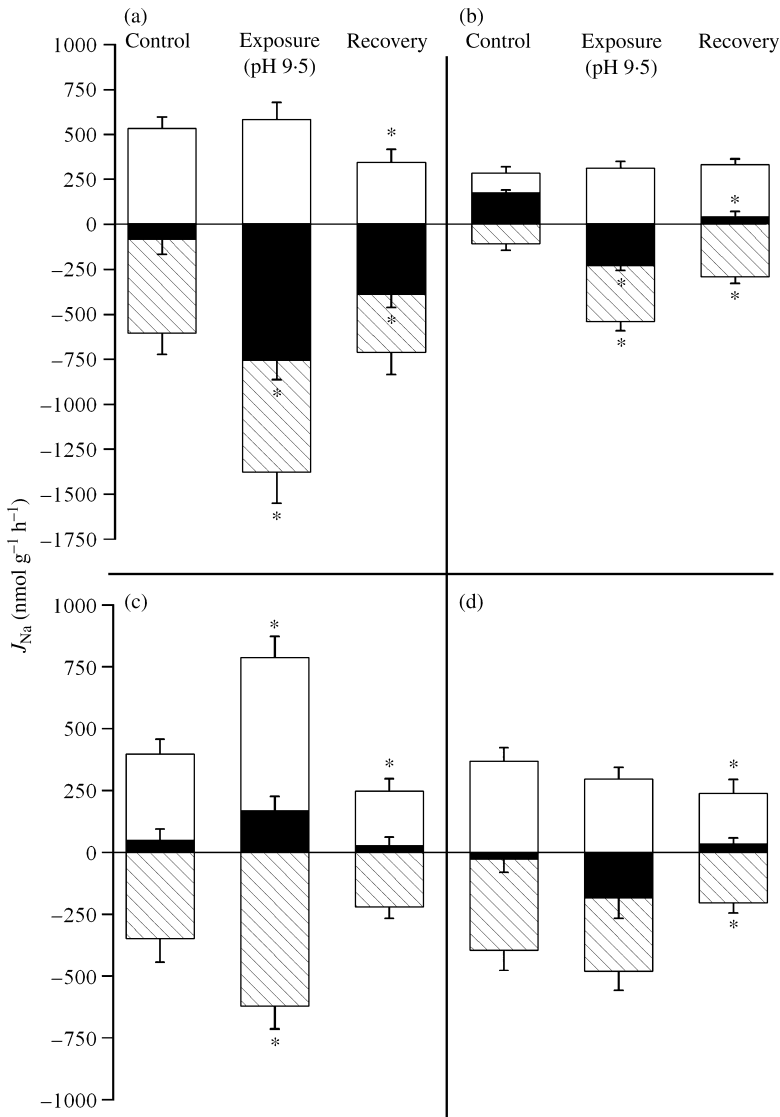


FIG. 2. Sodium fluxes [ $J_{Na}^{in}$  (□),  $J_{Na}^{net}$  (■) and  $J_{Na}^{out}$  (▨)] in (a) Slapton and (b) non-Slapton perch, and (c) rainbow and (d) brown trout before, during and after exposure to pH 9.5 water. Mean  $\pm$  s.e. values are shown with negative and positive values indicating a loss from or gain to the fish, respectively. \*, a significant ( $P < 0.05$ ) difference from the corresponding control value.

changes to  $Na^+$  fluxes were significant decreases in  $J_{Na}^{in}$  and  $J_{Na}^{out}$  in the recovery period.

Control rates of ammonia excretion varied  $>4.0$ -fold across all the species and sources of fishes ( $-181$  to  $-788$  nmol g<sup>-1</sup> h<sup>-1</sup>; Fig. 3). All groups showed a significant decrease in the rate of ammonia excretion during the experimental high pH period. The per cent change, compared to the relevant control, was lowest in non-Slapton roach (33.6%) and highest in brown trout (70.9%).

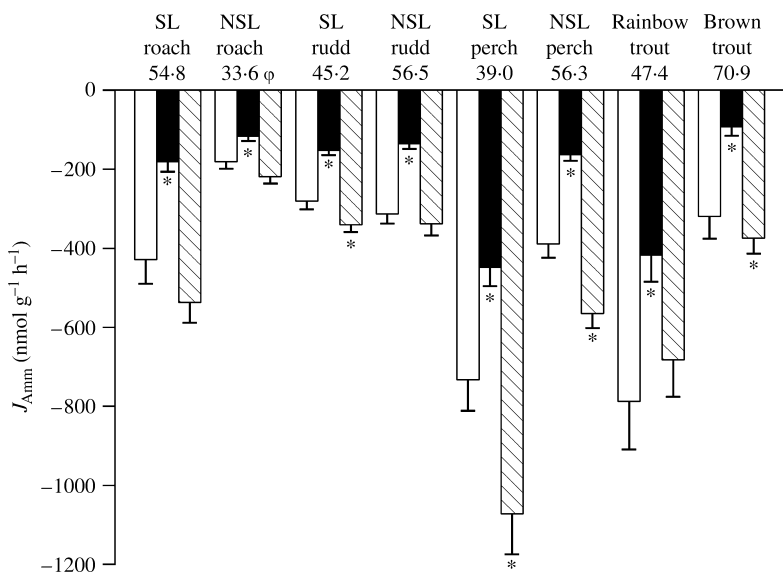


FIG. 3. Ammonia excretion rates in all fish groups (SL, Slapton Ley; NSL, non-Slapton Ley) before (□), during (■) and after (▨) exposure to pH 9.5 water. Values are means  $\pm$  s.e. \*, a significant ( $P < 0.05$ ) difference from the corresponding control value. The per cent decrease in ammonia excretion rates during pH 9.5 exposure compared to the corresponding control value are shown below the name of each fish group.  $\phi$ , a significant difference from the corresponding Slapton fish value.

The decrease in ammonia excretion experienced by non-Slapton roach was significantly less than the Slapton roach. Four groups of fishes showed a significant elevation in ammonia excretion, compared to control values, during the recovery period.

The  $\text{Na}^+$  uptake kinetics parameters,  $K_m$  and  $V_{\max}$ , for the different species are shown in Table I. Slapton perch had the lowest  $K_m$  (highest affinity) and Slapton rudd the highest  $K_m$  whilst non-Slapton rudd had the lowest  $V_{\max}$  and Slapton rudd the highest. Significant differences between the fishes from Slapton and non-Slapton sources were seen in two species of fishes (rudd and perch). The  $K_m$  and  $V_{\max}$  values for Slapton rudd were also significantly different from all other fish groups.

All three sizes of rainbow trout experienced a significant increase in  $\text{Na}^+$  efflux ( $J_{\text{Na}}^{\text{out}}$ ) during exposure to pH 9.5 (Table II). The small rainbow trout were the only group, which had a significant change (more negative) to net  $\text{Na}^+$  balance ( $J_{\text{Na}}^{\text{net}}$ ) during the experimental period. On the other hand, the medium and large fish experienced a significant increase in  $\text{Na}^+$  uptake ( $J_{\text{Na}}^{\text{in}}$ ) (1.8 and 1.7-fold respectively) during high pH exposure which balanced the increased  $J_{\text{Na}}^{\text{out}}$ . There were no significant relationships between fish mass and the  $\text{Na}^+$  fluxes or the per cent inhibition of ammonia excretion during high pH exposure. The  $\text{Na}^+$  uptake kinetics parameters,  $K_m$  and  $V_{\max}$ , for the different sizes of rainbow trout are shown in Table II. There were no significant differences between the kinetic parameters of the different sized trout.

TABLE I. Correlation coefficient ( $r^2$ ) of the Michaelis–Menten equation non-linear regression, affinity ( $K_m$ ) and capacity ( $V_{\max}$ ) for  $\text{Na}^+$  uptake values for the different fish groups (values are means  $\pm$  s.e.)

| Fish species  | $r^2$           |                 | $K_m$ ( $\mu\text{mol l}^{-1}$ ) |                  | $V_{\max}$ ( $\text{nmol g}^{-1} \text{h}^{-1}$ ) |                  |
|---------------|-----------------|-----------------|----------------------------------|------------------|---------------------------------------------------|------------------|
|               | Slapton Ley     | Non-Slapton Ley | Slapton Ley                      | Non-Slapton Ley  | Slapton Ley                                       | Non-Slapton Ley  |
| Roach         | $0.88 \pm 0.05$ | $0.89 \pm 0.04$ | $132.0 \pm 31.3$                 | $117.0 \pm 26.6$ | $536.4 \pm 87.2$                                  | $409.9 \pm 73.6$ |
| Rudd          | $0.92 \pm 0.03$ | $0.83 \pm 0.03$ | $468.5 \pm 78.5^{*\Phi}$         | $108.1 \pm 20.9$ | $774.2 \pm 81.3^{*\Phi}$                          | $303.1 \pm 20.9$ |
| Perch         | $0.89 \pm 0.05$ | $0.92 \pm 0.01$ | $72.8 \pm 14.2^*$                | $204.8 \pm 39.4$ | $626.7 \pm 89.7$                                  | $565.7 \pm 80.9$ |
| Rainbow trout | —               | $0.84 \pm 0.02$ | —                                | $121.8 \pm 25.0$ | —                                                 | $421.6 \pm 51.6$ |
| Brown trout   | —               | $0.87 \pm 0.09$ | —                                | $195.7 \pm 42.4$ | —                                                 | $506.3 \pm 62.2$ |

\*, a significant difference from the non-Slapton fish value;  $\Phi$ , a significant difference from all other values.

TABLE II.  $\text{Na}^+$  flux and ammonia excretion values for the three size groups of rainbow trout before (control), during (exp) and after (recovery) exposure to pH 9.5 water. The affinity ( $K_m$ ) and capacity ( $V_{\max}$ ) for  $\text{Na}^+$  uptake values for the different size groups of rainbow trout are shown as well as the correlation coefficient ( $r^2$ ) of the Michaelis–Menten equation non-linear regression. Mean  $\pm$  s.e. values are shown for the flux values with negative and positive values indicating a loss from or gain to the fish, respectively

|                                                                      | Small         |                 |               | Medium        |                 |               | Large         |                 |               |
|----------------------------------------------------------------------|---------------|-----------------|---------------|---------------|-----------------|---------------|---------------|-----------------|---------------|
|                                                                      | Control       | Exp             | Recovery      | Control       | Exp             | Recovery      | Control       | Exp             | Recovery      |
| $J_{\text{Na}}^{\text{in}}$ (nmol g <sup>-1</sup> h <sup>-1</sup> )  | 251 $\pm$ 39  | 285 $\pm$ 52    | 332 $\pm$ 43  | 410 $\pm$ 66  | 720 $\pm$ 66*   | 243 $\pm$ 59  | 166 $\pm$ 33  | 284 $\pm$ 37*   | 105 $\pm$ 29  |
| $J_{\text{Na}}^{\text{out}}$ (nmol g <sup>-1</sup> h <sup>-1</sup> ) | -189 $\pm$ 43 | -499 $\pm$ 64*  | -323 $\pm$ 48 | -340 $\pm$ 65 | -539 $\pm$ 51*  | -297 $\pm$ 86 | -152 $\pm$ 36 | -400 $\pm$ 29*  | -171 $\pm$ 38 |
| $J_{\text{Na}}^{\text{net}}$ (nmol g <sup>-1</sup> h <sup>-1</sup> ) | 62 $\pm$ 29   | -214 $\pm$ 17*  | 10 $\pm$ 18   | 70 $\pm$ 22   | 181 $\pm$ 86    | -54 $\pm$ 32  | 14 $\pm$ 23   | -116.1 $\pm$ 27 | -66 $\pm$ 23  |
| $J_{\text{Amm}}$ (nmol g <sup>-1</sup> h <sup>-1</sup> )             | -335 $\pm$ 45 | -43.6 $\pm$ 9*  | -322 $\pm$ 53 | -149 $\pm$ 16 | -36 $\pm$ 5*    | -150 $\pm$ 11 | -184 $\pm$ 19 | -20 $\pm$ 9*    | -188 $\pm$ 32 |
| $K_m$ ( $\mu\text{mol l}^{-1}$ )                                     |               | 86 $\pm$ 9      |               |               | 159 $\pm$ 34    |               |               | 117 $\pm$ 14    |               |
| $V_{\max}$ (nmol g <sup>-1</sup> h <sup>-1</sup> )                   |               | 632 $\pm$ 53    |               |               | 646 $\pm$ 61    |               |               | 587 $\pm$ 90    |               |
| $r^2$                                                                |               | 0.84 $\pm$ 0.02 |               |               | 0.82 $\pm$ 0.04 |               |               | 0.93 $\pm$ 0.02 |               |

\*, a significant ( $P < 0.05$ ) difference from the corresponding control value.

## DISCUSSION

The most important ionoregulatory parameter for the whole animal is the net flux of an ion (in this example  $J_{\text{Na}}^{\text{net}}$ ), which gives an indication of the general ion balance of the fish. One way to determine the relative tolerances of fishes to high pH is the extent to which  $J_{\text{Na}}^{\text{net}}$  is disturbed during exposure to alkaline water. Both Slapton and non-Slapton roach experienced a significant negative change to  $J_{\text{Na}}^{\text{net}}$  during exposure to alkaline water, but the change in  $J_{\text{Na}}^{\text{net}}$  was greater in the Slapton roach indicating that the non-Slapton roach were more tolerant of high pH. Disruption to ammonia excretion is also an important factor when considering tolerance to high pH, in severe cases accumulation of plasma ammonia can be lethal in fishes (Wright, 1995). Ammonia excretion was inhibited by 55% in Slapton roach but only 34% in the non-Slapton roach, which would also correlate with the  $J_{\text{Na}}^{\text{net}}$  data. Slapton and non-Slapton rudd showed a very similar pattern of  $\text{Na}^+$  fluxes following high pH exposure. Both groups experienced a small but significant negative change in  $J_{\text{Na}}^{\text{net}}$  during the recovery period. The non-Slapton rudd also experienced a slight decrease in  $J_{\text{Na}}^{\text{net}}$  during the experimental period, but remained in positive  $\text{Na}^+$  balance. Both rudd groups also experienced similar changes to ammonia excretion following alkaline exposure, indicating similar tolerances of Slapton and non-Slapton rudd to high pH. The  $\text{Na}^+$  flux data in the two groups of perch revealed a higher tolerance to high pH of the non-Slapton fish, with respect to net  $\text{Na}^+$  balance. On the other hand there was no statistical difference between the percent change in ammonia excretion for the two perch groups indicating comparable tolerances to high pH.

Some significant changes to unidirectional  $\text{Na}^+$  influx and efflux occurred in the recovery period with the two trout species but, surprisingly, neither showed any significant changes to net  $\text{Na}^+$  balance during or following high pH exposure. Brown trout in particular are generally thought to be very sensitive to changes in water quality (Davis, 1975; Brett & Blackburn, 1981; OECD, 1982; Eklöv *et al.*, 1999). Out of all the groups of fishes the brown trout showed the greatest inhibition of ammonia excretion during alkaline water exposure (70–90%), which would suggest they are sensitive to changes in water pH, despite the lack of ionoregulatory impairment.

Despite the fact that all of the fish groups experienced inhibition of ammonia excretion during exposure to pH 9.5 water, only four groups (out of eight) showed significant elevation of ammonia excretion upon return to fresh water in the recovery period. Elevation of plasma ammonia is inevitable if inhibition of excretion has occurred (Wilkie *et al.*, 1996; Scott *et al.*, 2005) and therefore when fishes are returned to fresh water a greater concentration gradient across the gill should support elevated excretion rates. Unfortunately there was a delay of *c.* 0.5 h following the experimental flux and before the recovery flux was started, during which time the chambers were flushed with clean water. It is likely that the majority of the accumulated plasma ammonia would have been excreted before the recovery flux had begun.

It must also be noted that these experiments only involved acute 1 h exposures which do not necessarily represent the situation that occurs in the wild in Slapton Ley, where fishes experience high pH water for several weeks or

even months during the summer. Nevertheless, these experiments still provide a convenient way to characterize the relative susceptibilities of the different species to high pH. Previous work on perch from Slapton Ley has in fact shown that during more chronic exposure to pH 9.5 net  $\text{Na}^+$  balance recovered fully by day 3 whereas ammonia excretion only partially recovered from 8 h onwards (Scott *et al.*, 2005).

A higher capacity for ion uptake rate ( $V_{\max}$ ) may be important in fishes that are more susceptible to disturbances in acid-base and ion balance (such as fishes in alkaline waters; Wilkie *et al.*, 1999), as it would allow more rapid recovery. Out of all the Slapton fish, however, rudd were the only group to have a significantly greater  $V_{\max}$  for  $\text{Na}^+$  uptake compared to the non-Slapton fish. When comparing all the fishes from Slapton and a non-Slapton sources there was no clear pattern with respect to  $\text{Na}^+$  uptake capacity. There were no trends with respect to  $K_m$  either; Slapton rudd had a significantly greater  $K_m$  (lower affinity) than the corresponding non-Slapton fish group, whereas Slapton perch had a significantly lower  $K_m$  (greater affinity) for  $\text{Na}^+$  uptake than the non-Slapton fish. Hence there is no evidence that fishes from Slapton have a greater affinity and capacity for  $\text{Na}^+$  uptake in order to cope with high pH exposure. On the basis of the net  $\text{Na}^+$  fluxes there is also no evidence that Slapton fishes have any greater tolerance to alkaline water as might be expected either through acclimatory changes during individual lifetimes, or selection over several generations.

The lack of inhibition of  $\text{Na}^+$  uptake, together with the increases in  $\text{Na}^+$  efflux experienced during five of the eight high pH exposures (including rainbow trout), qualitatively differs from previous studies on rainbow trout (Wright & Wood, 1985; Wilkie & Wood, 1994; Wilkie *et al.*, 1999; Laurent *et al.*, 2000). Furthermore, in two groups within the present study (non-Slapton roach and rainbow trout)  $\text{Na}^+$  influx actually increased during high pH exposure. The reasons for such differences between the current and previous studies are not clear and a number of factors could be involved including differences in experimental conditions and species physiology. The length of the flux period was longer also (3 h, compared to 1 h) in the previous studies, and therefore the initial (within the first hour) responses of fish to high pH could have been masked by subsequent changes to the  $\text{Na}^+$  fluxes.

An increase in  $\text{Na}^+$  efflux during high pH exposure has not been reported previously, and although the mechanisms involved are not known some authors have suggested that  $\text{Ca}^{2+}$  levels in the water may be an important factor in controlling branchial permeability during high pH, as it is during acid exposure (McDonald, 1983; Freda & McDonald, 1988; Yesaki & Iwama, 1992). Experiments conducted with Slapton perch exposed to pH 9.5 in water with low ( $[\text{Na}^+] = 289$ ,  $[\text{Ca}^{2+}] = 416 \mu\text{mol l}^{-1}$ ) or high ( $[\text{Na}^+] = 633$ ,  $[\text{Ca}^{2+}] = 643 \mu\text{mol l}^{-1}$ ) ion levels demonstrated a positive effect of water ion concentration on limiting  $\text{Na}^+$  efflux (3.0 and 2.0-fold increases in  $J_{\text{Na}}^{\text{out}}$  during high pH in low and high ion water, respectively; pers. obs.). It therefore seems that water  $\text{Ca}^{2+}$  (and  $\text{Na}^+$ ) is an important factor in controlling gill permeability during high pH exposure, in perch at least. In the previous work with rainbow trout the average ambient  $\text{Ca}^{2+}$  concentration was higher at c.  $1000 \mu\text{mol l}^{-1}$ , whereas in the current series of experiments it averaged  $620 \mu\text{mol l}^{-1}$ . This reduced

availability of external  $\text{Ca}^{2+}$  may help explain why the fishes in the present study were more susceptible to increased  $\text{Na}^+$  efflux following high pH exposure.

This increased  $\text{Na}^+$  efflux during exposure to pH 9.5 water only occurred in some of the groups of fishes: rainbow trout and Slapton and non-Slapton rudd and roach. It therefore seems that the ability to resist changes in branchial permeability during alkaline exposure is species specific. Similarly, acid tolerance in fishes has been linked with an ability to control branchial permeability (Freda & McDonald, 1988; McDonald *et al.*, 1991). In particular interspecific differences in the levels of calmodulin, a  $\text{Ca}^{2+}$  binding protein, in gill mucus and hence branchial  $\text{Ca}^{2+}$  binding activity are thought to be related to acid tolerance (Flik *et al.*, 1984; Freda & McDonald, 1988). It is therefore possible that the differences in the ability of the species in this study to control  $\text{Na}^+$  efflux during exposure to alkaline water is due to differences in branchial  $\text{Ca}^{2+}$  binding.

The explanation for the unexpected increase in  $\text{Na}^+$  influx in two groups of fishes during exposure to high pH water is less clear. In theory raising the water pH should indeed favour  $\text{Na}^+$  uptake, with the increased  $\text{H}^+$  gradient across the apical gill membrane favouring  $\text{H}^+$  extrusion and hence  $\text{Na}^+$  uptake, which is thought to be linked with proton excretion (Potts, 1994). Indeed the opposite of this is thought to occur under acid conditions where the electrochemical gradient operates against the proton pump and hence inhibits  $\text{Na}^+$  uptake (McDonald, 1983). The reason why stimulation of  $\text{Na}^+$  influx occurred during this but not other studies, could be due to differences in the buffering capacity of the experimental water, best estimated as TALK, which was almost double in the previous rainbow trout work ( $2.1 \text{ mmol l}^{-1}$ ; Wright & Wood, 1985; Wilkie & Wood, 1994; Wilkie *et al.*, 1999; Laurent *et al.*, 2000) compared to the current study ( $1.0 \text{ mmol l}^{-1}$ ). This means that in the previous experiments the gill boundary layer pH will have been closer to the measured bulk water pH and may have been high enough to cause non-competitive inhibition of the ion transporters involved in  $\text{Na}^+$  uptake, which outweighed any competitive stimulation of  $\text{Na}^+$  uptake due to an enhanced  $\text{H}^+$  excretion gradient. This argument is speculative but in agreement McGeer & Eddy (1998) found that buffering of the experimental water reduced the tolerance of rainbow trout to alkaline water with significant decreases in plasma ions.

Little is known about the physiology of the Slapton fish species and a number of  $\text{Na}^+$  uptake mechanisms have previously been suggested to occur in teleosts (Evans, 1980; McDonald & Prior, 1988; Avella & Bornancin, 1989; Salama *et al.*, 1999; Kelly & Wood, 2001; Preest *et al.*, 2005). The varying effect that high pH water had on these different groups of fishes with respect to their  $\text{Na}^+$  fluxes suggests that there may well be different  $\text{Na}^+$  uptake mechanisms operating in these species, which may explain why only two groups experienced stimulation of  $\text{Na}^+$  influx during high pH exposure. There are also fishes, such as those that inhabit the acidic waters of the Amazon, which possess  $\text{Na}^+$  uptake mechanisms which are insensitive to pHs as low as 3.5 (Gonzalez & Wilson, 2001; Gonzalez *et al.*, 2002). It could therefore be plausible that the fishes used in the present study have  $\text{Na}^+$  uptake mechanisms that are insensitive to higher pHs.

It has been previously reported that the body mass of freshwater organisms (fishes and crustaceans) is inversely correlated with the rates of both branchial  $\text{Na}^+$  uptake and efflux (Bianchini *et al.*, 2002; Grosell *et al.*, 2002). In the current study, however, there was no significant relationship between body mass and  $\text{Na}^+$  fluxes during control or alkaline conditions in the three sizes of rainbow trout. It is likely that, over the relatively small size range in the current study (2–40 g) this relationship is undetectable. Despite the differences in the control ammonia excretion rates in the three size groups of rainbow trout they showed remarkably similar levels of per cent inhibition during pH 9.5 exposure. Therefore, it can be concluded that the range of fish sizes used in the current study is not an important factor in explaining any differences found in their measured responses to alkaline conditions. Similarly the kinetics of  $\text{Na}^+$  uptake should not have been affected by differences in fish mass.

Excellent technical assistance from J. Shears and P. Shears is greatly appreciated. We would like to thank the Whitely Wildlife Conservation Trust, The Environment Agency and South West Water for their financial support.

### References

- Avella, M. & Bornancin, M. (1989). A new analysis of ammonia and sodium transport through the gills of the fresh water rainbow trout (*Salmo gairdneri*). *Journal of Experimental Biology* **142**, 155–175.
- Bianchini, A., Grosell, M., Gregory, S. M. & Wood, C. M. (2002). Acute silver toxicity in aquatic animals is a function of sodium uptake rate. *Environmental Science & Technology* **36**, 1763–1766.
- Brett, J. R. & Blackburn, J. M. (1981). Oxygen requirements for growth of young coho (*Oncorhynchus kisutch*) and sockeye (*Oncorhynchus nerka*) salmon at 15°C. *Canadian Journal of Fisheries and Aquatic Sciences* **38**, 399–404.
- Brett, J. R. & Zala, C. A. (1975). Daily pattern of nitrogen excretion and oxygen consumption of sockeye salmon (*Oncorhynchus nerka*) under controlled conditions. *Journal of the Fisheries Research Board of Canada* **32**, 2479–2486.
- Davis, J. C. (1975). Minimal dissolved oxygen requirements of aquatic life with emphasis on Canadian species: a review. *Journal of the Fisheries Research Board of Canada* **32**, 2295–2332.
- Eklöv, A. G., Greenberg, L. A., Bronmark, C., Larsson, P. & Berglund, O. (1999). Influence of water quality, habitat and species richness on brown trout populations. *Journal of Fish Biology* **54**, 33–43. doi: 10.1111/j.1095-8649.1999.tb00610.x
- Evans, D. H. (1980). Kinetic studies of ion transport by fish gill epithelium. *American Journal of Physiology* **238**, R224–R230.
- Flik, G., Van Rijs, J. H. & Wendelaar Bonga, S. E. (1984). Evidence for the presence of calmodulin in fish mucus. *European Journal of Biochemistry* **138**, 651–654.
- Freda, J. & McDonald, D. G. (1988). Physiological correlates of interspecific variation in acid tolerance in fish. *Journal of Experimental Biology* **136**, 243–258.
- Gonzalez, R. J. & Wilson, R. W. (2001). Patterns of ion regulation in acidophilic fish native to the ion-poor, acidic Rio Negro. *Journal of Fish Biology* **58**, 1680–1690. doi: 10.1111/j.1095-8649.2001.tb02322.x
- Gonzalez, R. J., Wilson, R. W., Wood, C. M., Patrick, M. L. & Val, A. L. (2002). Diverse strategies for ion regulation in fish collected from the ion-poor, acidic Rio Negro. *Physiological and Biochemical Zoology* **75**, 37–47.
- Goss, G. G. & Wood, C. M. (1990).  $\text{Na}^+$  and  $\text{Cl}^-$  uptake kinetics, diffusive effluxes and acidic equivalent fluxes across the gills of rainbow trout. 1. Responses to environmental hyperoxia. *Journal of Experimental Biology* **152**, 521–547.

- Goss, G. G. & Wood, C. M. (1991). Two-substrate kinetic analysis: a novel approach linking ion and acid-base transport at the gills of freshwater trout, *Oncorhynchus mykiss*. *Journal of Comparative Physiology B* **161**, 635–646.
- Goss, G. G., Perry, S. F., Wood, C. M. & Laurent, P. (1992). Mechanisms of ion and acid-base regulation at the gills of freshwater fish. *Journal of Experimental Zoology* **263**, 143–159.
- Grosell, M., Nielsen, C. & Bianchini, A. (2002). Sodium turnover rate determines sensitivity to acute copper and silver exposure in freshwater animals. *Comparative Biochemistry and Physiology C* **133**, 287–303.
- Hughes, G. M. (1966). The dimensions of fish gills in relation to their function. *Journal of Experimental Biology* **45**, 177–195.
- Johnes, P. J. & Heathwaite, A. L. (1997). Modelling the impact of land use change on water quality in agricultural catchments. *Hydrological Processes* **11**, 269–286.
- Kelly, S. P. & Wood, C. M. (2001). The cultured branchial epithelium of the rainbow trout as a model for diffusive fluxes of ammonia across the fish gill. *Journal of Experimental Biology* **204**, 4115–4124.
- Laurent, P., Wilkie, M. P., Chevalier, C. & Wood, C. M. (2000). The effect of highly alkaline water (pH 9.5) on the morphology and morphometry of chloride cells and pavement cells in the gills of the freshwater rainbow trout: relationship to ionic transport and ammonia excretion. *Canadian Journal of Zoology* **78**, 307–319.
- Maetz, J. (1956). Les échanges de sodium chez le poisson *Carassius auratus* L. Action d'un inhibiteur de l'anhydrase carbonique. *Journal of Physiology* **48**, 1085–1099.
- Marshall, W. S. & Grosell, M. (2005). Ion transport, osmoregulation, and acid-base balance. In *The Physiology of Fishes*, 3rd edn (Evans, D. H., ed.), pp. 177–230. Boca Raton, FL: CRC Press.
- McDonald, D. G. (1983). The effects of  $H^+$  upon the gills of freshwater fish. *Canadian Journal of Zoology* **61**, 691–703.
- McDonald, D. G. & Prior, E. T. (1988). Branchial mechanisms of ion and acid-base regulation in the fresh water rainbow trout, *Salmo gairdneri*. *Canadian Journal of Zoology* **66**, 2699–2708.
- McDonald, D. G. & Wood, C. M. (1981). Branchial and renal acid and ion fluxes in the rainbow trout, *Salmo gairdneri*, at low environmental pH. *Journal of Experimental Biology* **93**, 101–118.
- McDonald, D. G., Cavdek, V. & Ellis, R. (1991). Gill design in freshwater fishes—interrelationships among gas-exchange, ion regulation, and acid-base regulation. *Physiological Zoology* **64**, 103–123.
- McGeer, J. C. & Eddy, F. B. (1998). Ionic regulation and nitrogenous excretion in rainbow trout exposed to buffered and unbuffered freshwater of pH 10.5. *Physiological Zoology* **71**, 179–190.
- OECD (1982). *Eutrophication of Waters, Monitoring Assessment and Control*. Paris: Organisation for Economic Co-operation and Development.
- Paerl, H. W., Fulton, R. S. & Moisaner, P. H. (2001). Harmful freshwater algal blooms, with an emphasis on cyanobacteria. *The Scientific World* **1**, 76–113.
- Potts, W. T. W. (1994). Kinetics of sodium uptake in fresh water animals—a comparison of ion-exchange and proton pump hypotheses. *American Journal of Physiology* **266**, R315–R320.
- Preest, M. R., Gonzalez, R. J. & Wilson, R. W. (2005). A pharmacological examination of  $Na^+$  and  $Cl^-$  transport in two species of freshwater fish. *Physiological and Biochemical Zoology* **78**, 259–272.
- Randall, D., Burggren, W. & French, K. (Eds) (2002). *Eckert Animal Physiology*. New York: W.H. Freeman and Company.
- Salama, A., Morgan, I. J. & Wood, C. M. (1999). The linkage between  $Na^+$  uptake and ammonia excretion in rainbow trout: kinetic analysis, the effects of  $(NH_4)_2SO_4$  and  $NH_4HCO_3$  infusion and the influence of gill boundary layer pH. *Journal of Experimental Biology* **202**, 697–709.

- Schmidt-Nielsen, K. (1984). *Scaling: Why is Animal Size so Important?* 1st edn. Cambridge: Cambridge University Press.
- Scott, D. M. (2003). The physiology and behaviour of fish from a freshwater eutrophic lake, Slapton Ley, Devon. PhD Thesis, University of Exeter.
- Scott, D. M., Lucas, M. C. & Wilson, R. W. (2005). The effect of high pH on ion balance, nitrogen excretion and behaviour in freshwater fish from an eutrophic lake: a laboratory and field study. *Aquatic Toxicology* **73**, 31–43.
- Verdouw, H., Van Echteld, C. J. A. & Dekkers, E. M. J. (1978). Ammonia determination based on indophenol formation with sodium salicylate. *Water Research* **12**, 399–402.
- Wilkie, M. P. & Wood, C. M. (1991). Nitrogenous waste excretion, acid-base regulation, and ionoregulation in rainbow trout (*Oncorhynchus mykiss*) exposed to extremely alkaline water. *Physiological Zoology* **64**, 1069–1086.
- Wilkie, M. P. & Wood, C. M. (1994). The effects of extremely alkaline water (pH 9.5) on rainbow trout gill function and morphology. *Journal of Fish Biology* **45**, 87–98. doi: 10.1111/j.1095-8649.1994.tb01288.x
- Wilkie, M. P., Wright, P., Iwama, G. K. & Wood, C. M. (1993). The physiological responses of the Lahontan cutthroat trout (*Oncorhynchus clarki henshawi*), a resident of highly alkaline pyramidal lake (pH 9.4), to challenge at pH 10. *Journal of Experimental Biology* **175**, 173–194.
- Wilkie, M. P., Simmons, H. E. & Wood, C. M. (1996). Physiological adaptations of rainbow trout to chronically elevated water pH (pH = 9.5). *Journal of Experimental Zoology* **274**, 1–14.
- Wilkie, M. P., Laurent, P. & Wood, C. M. (1999). The physiological basis for altered  $\text{Na}^+$  and  $\text{Cl}^-$  movements across the gills of rainbow trout (*Oncorhynchus mykiss*) in alkaline (pH = 9.5) water. *Physiological and Biochemical Zoology* **72**, 360–368.
- Wilson, R. W., Wright, P. M., Munger, S. & Wood, C. M. (1994). Ammonia excretion in fresh water rainbow trout (*Oncorhynchus mykiss*) and the importance of gill boundary-layer acidification—lack of evidence for  $\text{Na}^+/\text{NH}_4^+$  exchange. *Journal of Experimental Biology* **191**, 37–58.
- Wilson, J. M., Iwata, K., Iwama, G. K. & Randall, D. J. (1998). Inhibition of ammonia excretion and production in rainbow trout during severe alkaline exposure. *Comparative Biochemistry and Physiology* **121**, 99–109.
- Wood, C. M., Wilson, P., Bergman, H. L., Bergman, A. N., Laurent, P., Otiang'a-Owiti, G. & Walsh, P. J. (2002). Ionoregulatory strategies and the role of urea in the Magadi tilapia (*Alcolapia grahami*). *Canadian Journal of Zoology* **80**, 503–515.
- Wright, P. A. (1995). Nitrogen excretion—3 end-products, many physiological roles. *Journal of Experimental Biology* **198**, 273–281.
- Wright, P. A. & Wood, C. M. (1985). An analysis of branchial ammonia excretion in the fresh water rainbow trout—effects of environmental pH change and sodium uptake blockade. *Journal of Experimental Biology* **114**, 329–353.
- Wright, P. A., Randall, D. J. & Perry, S. F. (1989). Fish gill water boundary layer—a site of linkage between carbon dioxide and ammonia excretion. *Journal of Comparative Physiology B* **158**, 627–635.
- Yesaki, T. Y. & Iwama, G. K. (1992). Survival, acid-base regulation, ion regulation, and ammonia excretion in rainbow trout in highly alkaline hard water. *Physiological Zoology* **65**, 763–787.